Interactive Learning Platform

THE PROBLEM

- **Passive Learning Trap:** Learners watch hours of content but can't apply knowledge.
- **No Accountability:** No checks if learners are actually paying attention.
- **Disconnected Practice:** Switching between video platform and editor breaks flow.



Our Solution

> **A gamified platform where learners MUST actively participate to progress.**

- **Active Learning:** Forces engagement through mandatory quizzes.
- **Integrated Environment:** Code practice embedded directly in the lesson player.
- **Visual Growth:** Tracks progress with trees, streaks, and badges.



Platform Overview

Learn

Video lessons organized by modules with real-time tracking.

Quiz

Gamified quizzes with lives & timers. Must pass to continue.

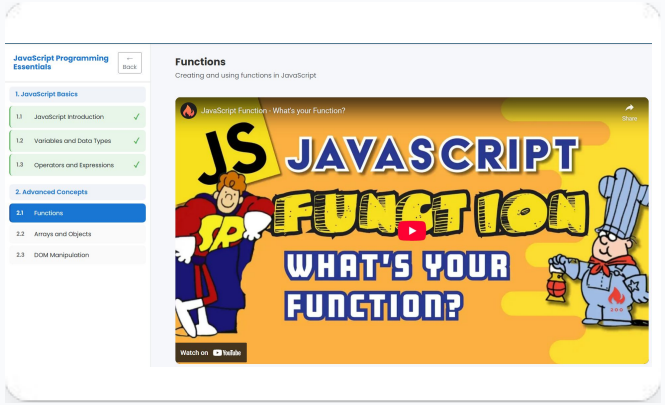
Practice

Built-in code editor (Monaco) supporting HTML, CSS, & JS.



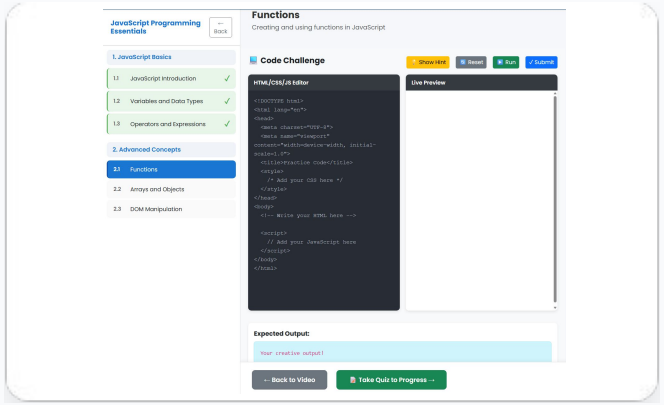
React TypeScript Node.js MongoDB

The Learning Experience



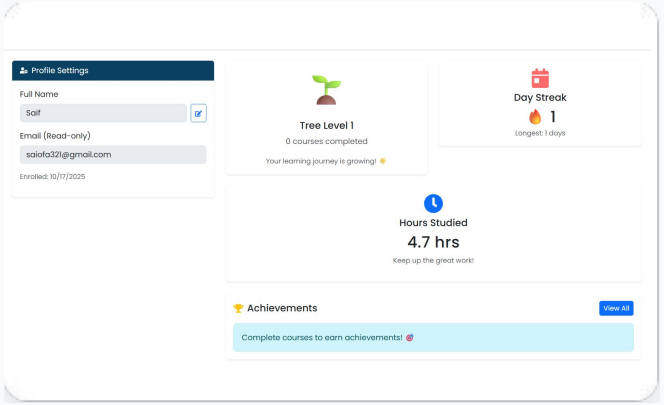
Lesson Player

Sidebar navigation, video integration, and progress tracking.



Code Editor

Integrated Monaco editor with live preview for instant practice.



Profile

Hours studied, tree level, and day streaks.

User Journey

Enroll

Register, browse the catalog, and start your first module.

Learn & Quiz

Watch content, then pass the mandatory quiz.

Next Lesson

Progress unlocks automatically upon success.

Complete

Grow your tree, earn badges, and maintain your streak.

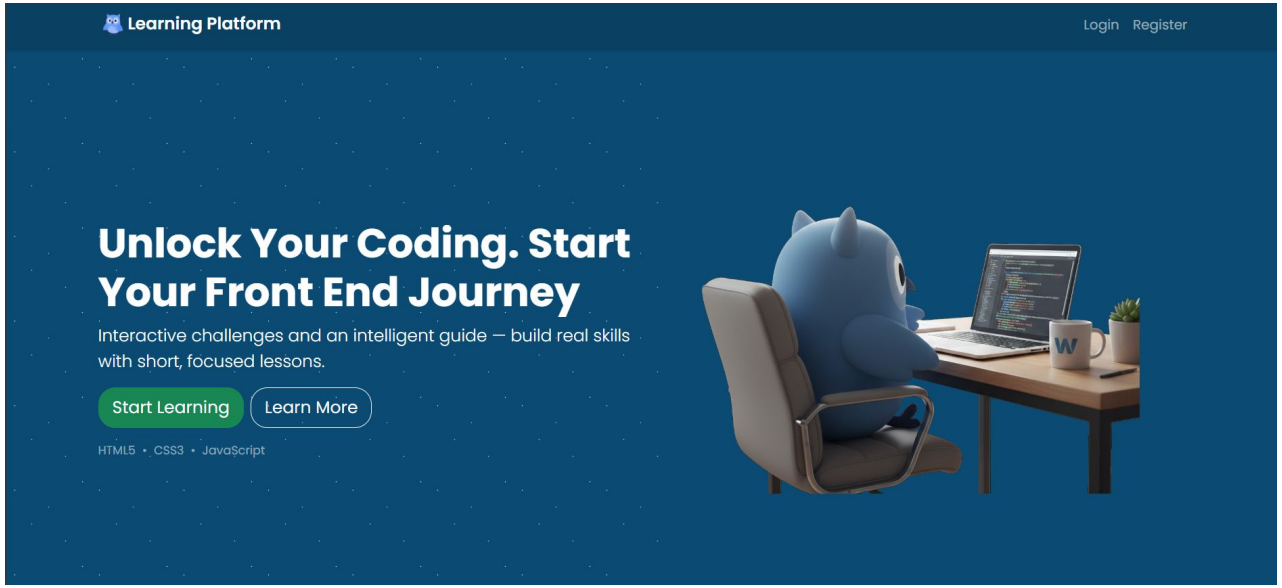
First Impressions: Landing page

Hero Section

Engaging headline and clear CTA buttons ("Start Learning", "Browse Courses") to drive conversion.

Feature Highlights

Clean layout showcasing gamification and coding features to new visitors.

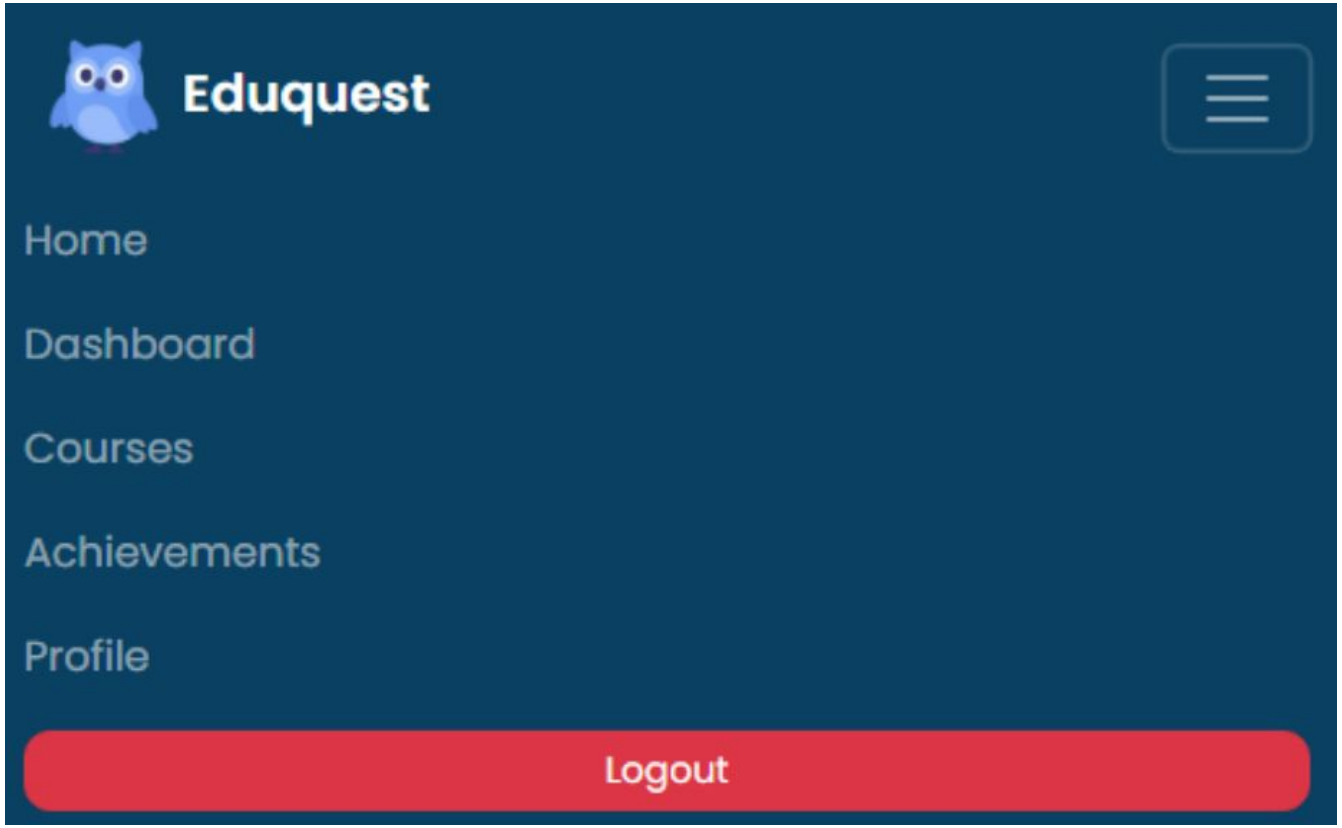


the Navbar:

- **Navigation Bar**
  Responsive design with a persistent profile dropdown, course search, and quick links.

- Clear icons and labels that stay readable even on compact displays.

- Fully responsive layout that adapts smoothly without breaking or crowding elements.



Celebrating Success: Achievements

Badge Collection

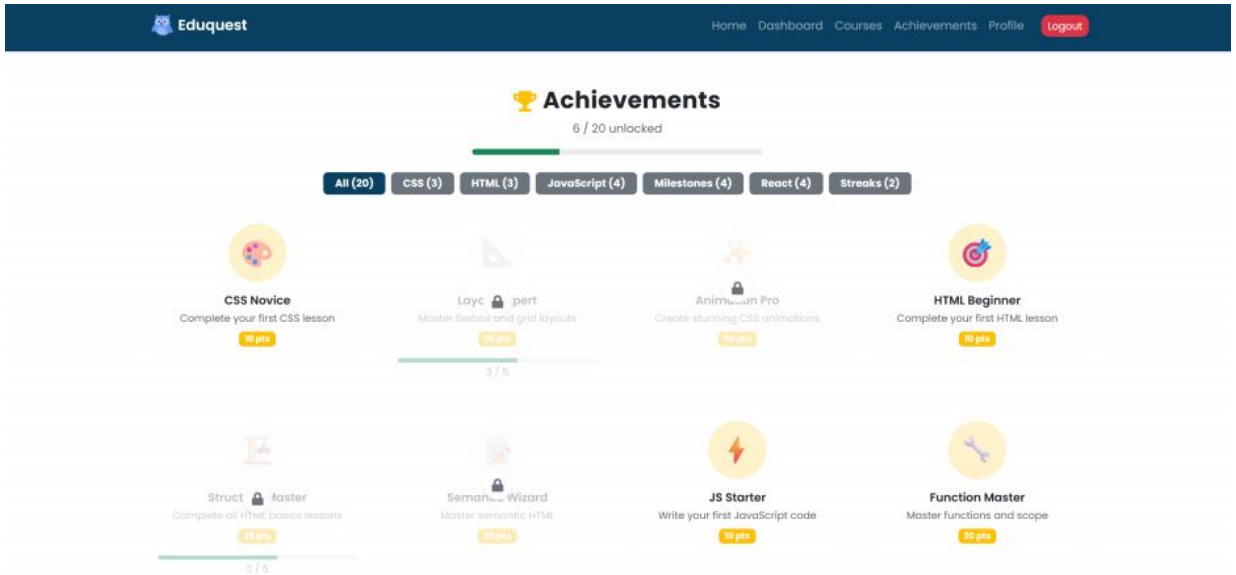
A visual grid of all earnable badges, motivating users to unlock them all.

Rarity Levels

Distinct visual styles for Common, Rare, and Epic achievements to signal prestige.

Social Sharing

One-click options to share earned badges on social media, boosting platform visibility.



Dashboard Overview:

Centralized Learning Hub:

The dashboard provides a clear and organized view of the user's learning progress. It displays courses dynamically, making it easy for users to track what they are currently studying.

Dynamic Course Display:

Courses are fetched from the backend and shown in clean, modern cards. Each card includes key details such as course name, description, progress, and a quick action button, helping users continue learning easily.

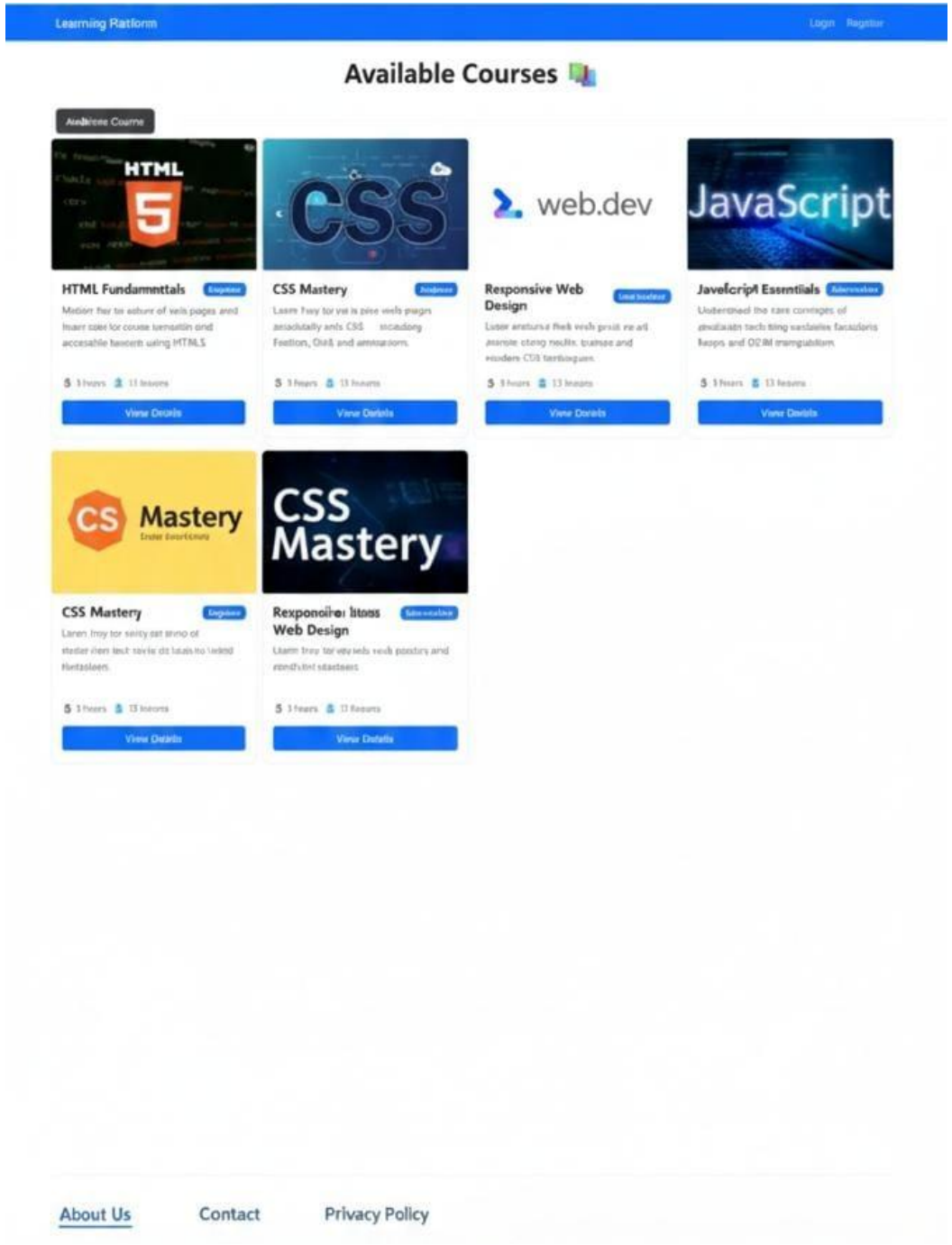
Responsive, User-Friendly Design

Using Bootstrap's grid system and modern UI elements, the dashboard adapts smoothly to all screen sizes. Icons and layouts enhance readability and make navigation simple and intuitive.

Courses Page Overview

- The page displays all courses in an organized, dynamic layout using a reusable Card component. Each card presents key course details clearly and allows quick navigation to the full course page.

- Add Course Functionality
  An “Add New Course” button directs users to the course creation form, enabling easy and efficient course management.



Add Courses Page

Secure & Validated Form

The Add Course page includes a form with full validation using Zod. Each field (name, image URL, level, duration, lessons, category, and description) is validated before submitting to ensure accuracy and prevent errors.

Course Creation Functionality

When the form is submitted, the new course is sent to the server using Axios. If successful, a success message appears using React-Toastify, and the user is redirected back to the Courses page.

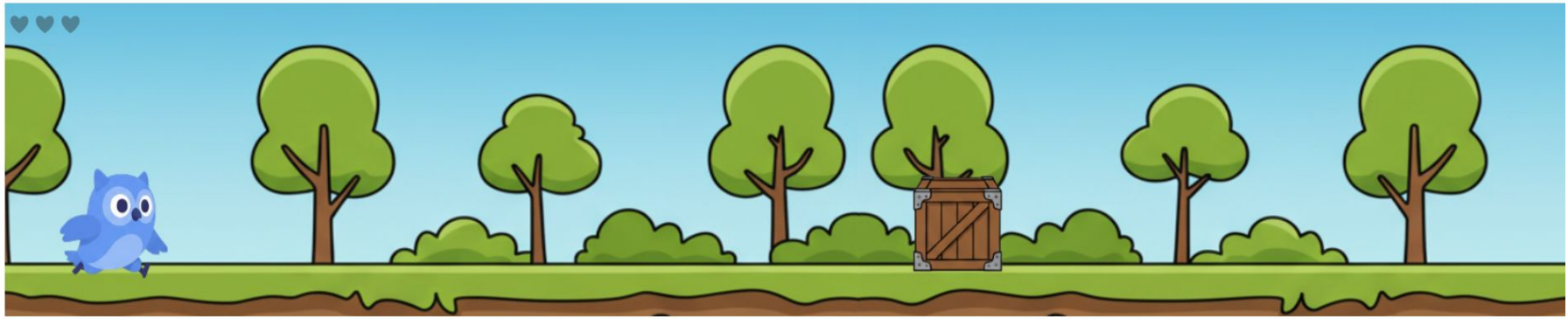
User-Friendly Layout

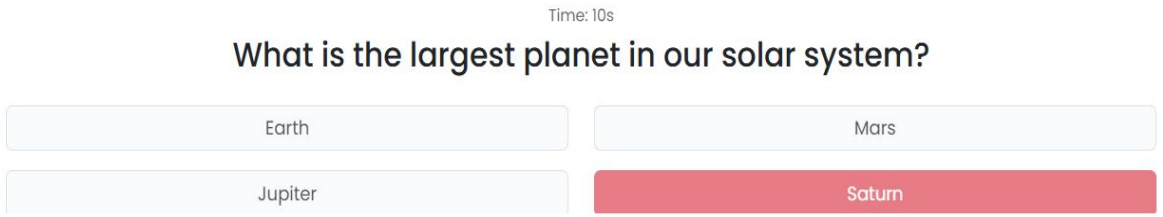
The form appears inside a styled card with labeled inputs and error messages under each field, making it easy for users to add new courses with confidence and it’s easy to write in it..

Game Overview

Quizzes as games
to make studying more fun

Trials
to learn from your mistakes





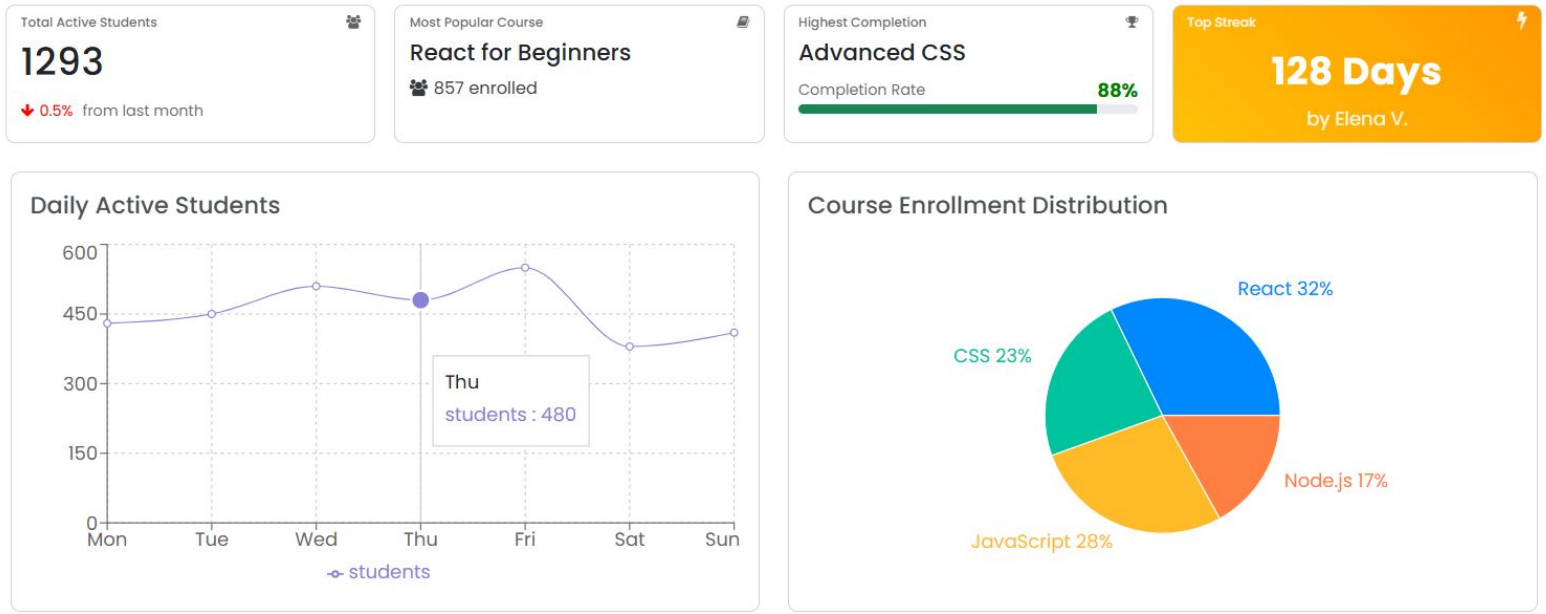
Admin dashboard

Platform Analytics

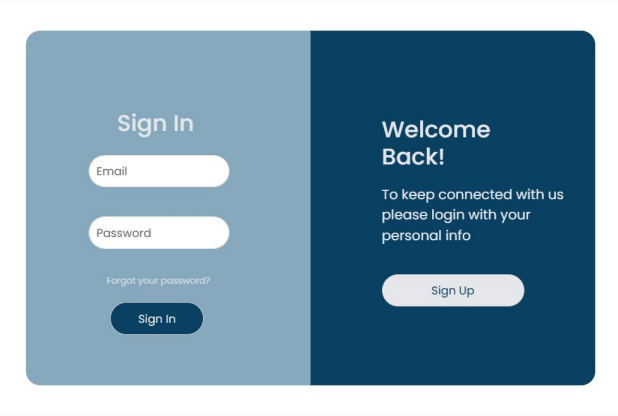
Key metrics on course completion, quiz pass/fail rates, and overall user engagement data.

System Status

Real-time data on server uptime and platform performance

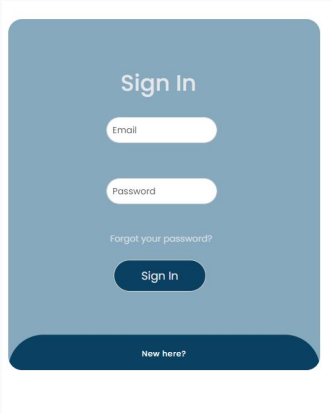


Login page Overview



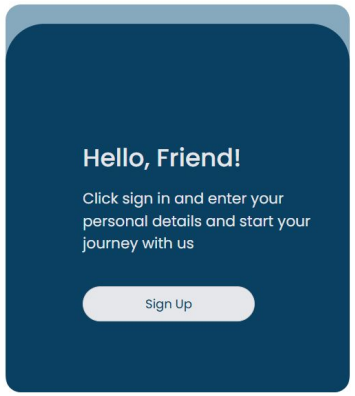
Seamless Onboarding

Fast, secure sign-in/sign-up process to get users learning quickly.



Simple And Clean UI

Minimalist design to reduce friction and focus the user on access.



Adaptive Layout

Ensures the entire platform adapts seamlessly to all screen sizes, from mobile phones to large desktop monitors.

What's Next

- 🤖 **AI Tutoring**
  Real-time help while learning.
- 📱 **Mobile App**
  Learn on the go with React Native.
- 👥 **Peer Learning**
  Study groups and pair programming.



"We transformed passive video-watching into active, gamified learning."

Questions?